Comparison of tribological performance of some neat polymer and polymers composites

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In this study, the tribological performances of some neat polymers and graphite and wax filled polymer composites were examined, compared and evaluated. The investigated neat polymers are polyoxymethylene, ultrahigh molecular weight polyethylene and polyamide (PA6) engineering polymers. The studied composites are 10 wt-% graphite and 6 wt-% wax filled PA6 polymer. Wear examination was performed on pin on disc arrangement. The test conditions were atmospheric, sliding speed range of 0·5–2·0 m s^{-1} and contact pressure range of 1·77–5·31 MPa. Furthermore, scanning and optical examinations of worn surfaces were carried out and considered. The results show the insignificant variation in polymer tribological performance with the increase in combined speed and pressure factor values. It was concluded that the PA6 + 10 wt-% graphite composite showed the highest tribological performance and suggested its application at high speed environment. Finally, it is also concluded that the wear mechanism of the worn surfaces includes transfer film, deformation and adhesive processes.

Keywords: Polymer, Wax, Composite, POM

Introduction

Polymer and polymer composites have been increasingly used in various industrial applications, such as aerospace, automotive, railway and chemical industries, because these polymers provide self-lubricant conditions and high strength/weight ratio in comparison to classic material. Among these polymers and polymer composites are neat polyoxymethylene (POM), ultra high molecular weight polyethylene (UHMWPE), polyamide (PA6) and PA6 composites.[1]

Polyoxymethylene is a high cost, rigid polymer with low moisture absorption, high strength, very high surface temperature limits, low creep, good electrical characteristics, transparency, self-extinguishing ability and high resistance to grease and chemicals.[2,3] Mergler *et al.*[4] studied the material transfer of POM in sliding contact and concluded that the transition of POM to the metal counter surface leads to an increase in the coefficient of friction. Samyn *et al.*[5–7] and Endos and Marui[8] studied the friction and wear of POM and concluded the influences of geometry and load scale on the POM transition mechanism. Mao *et al.*[9] studied the friction and wear of POM, and Chen *et al.*[10] investigated the wear performance of POM/polytetrafluoroethylene (PTFE) blends. They concluded that POM appears in wear by thermal softening and melting of worn surface, and the surface temperature rise is important particularly under high load application.

Ultrahigh molecular weight polyethylene is a useful thermoplastic polymer for engineering applications. This is due to its excellent properties, such as chemical stability, high impact strength, high wear resistance and low friction.[11,12] On the other hand, UHMWPE has difficult processing route, high processing cost, low hardness and creep resistance. The low hardness and low creep resistance may lead to excessive permanent deformation of the bearing surface and ultimately affect the surface geometry.

PA6 is an engineering plastic used in electrical/electronics, automobile, packaging, textiles and consumer applications because of their excellent mechanical properties.[13,14] However, limitations in mechanical properties level, low heat deflection temperature, high water absorption and dimensional instability of neat (pure) PA6 have prevented its wide range of applications. Hence, solid lubricants and graphite were added to polyamide 6 to widen and increase its application range. Graphite and wax filled PA6 have found its application field in high velocity applications without the need for exterior lubrication.[15] Samyn *et al.* investigated the softening and melting mechanisms of polyamides under adhesive conditions and concluded that the transfer film formed under continuous softening provides high friction state.[16] Basavaraj *et al.*[17] and Xing *et al.*[18] studied the influence of the addition of PTFE on the tribological behaviour of PA6 composites and recorded the enhancement in wear rate of PA6 composites. Wang and Gu[19] and Li and Cheng[20] studied the influence of carbon fibre filler on the tribological behaviour of PA6 and concluded a large improvement

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in tribological performance under an oil lubricant environment. Li *et al.*[21] studied the wear and friction of PA6 filled with PTFE and UHMWPE and concluded a decrease in wear rate with the increasing applied load.

In general, the tribological performance of polymers is influenced by material combination, environment and test conditions (applied load and sliding speed). Franke *et al.*[22] reported that the friction coefficient of polymers and polymer composites can be reduced and the wear resistance increased by selecting the right material combinations. Unal and Mimaroglu[23,24] studied the individual influence of applied load and sliding speed on the tribological performance of polymer and polymer composite. They concluded their high influences on friction coefficient and wear rate values of polymer materials.

The main aims of this study include examination, evaluation and comparison of the tribological performance of POM, UHMWPE, PA6 neat polymers and graphite and wax filled PA6 composites under a range of sliding speeds and applied load values individually and as combined speed and pressure factor (PV) range. The results show the insignificant variation in tribological performance with the increase in PV values. It was concluded that among all the tested neat polymers and polymer composites of this investigation, the PA6+10 wt-% graphite composite showed the highest surface tribological performance and suggested its application at high speed environment. Finally, it is also concluded that the wear mechanism of the worn surfaces includes transfer film, deformation and adhesive processes.

Experimental

During this investigation, the wear tests were carried out using an in house designed pin on disc wear test rig. The test rig consists of a stainless steel table that is mounted on a turntable, a variable speed motor that provides unidirectional motion to the turntable and hence to the steel disc sample and a pin sample holder that is rigidly attached to a pivoted loading arm. This loading arm is supported in bearing arrangements to allow loads to be applied to the specimen. Neat polymer samples were prepared by injection moulding process. The composite samples were made by extrusion followed by injection moulding processes. The materials, material suppliers and test conditions of the experimental tests are given in Tables 1 and 2. To examine the tribological performance of the materials, cylindrical pin specimens of size 6 mm diameter and 50 mm length of POM, PA6, UHMWPE and PA6 graphite and wax filled composites were tested against a stainless steel disc. During the test, friction force was measured at 100 readings per second rate using a load cell, and the average value for the friction force readings was taken and considered. The sliding wear data reported here are the average of at least three runs. The average mass loss was used to calculate the specific wear rate K_0 ($m^3\ N^{-1}\ m^{-1}$) as shown below

$$K_0 = \Delta m / LF\rho$$

where Δm is the average mass loss, L is the sliding distance, F is the applied load (N) and ρ is the density of the materials.

Results and discussion

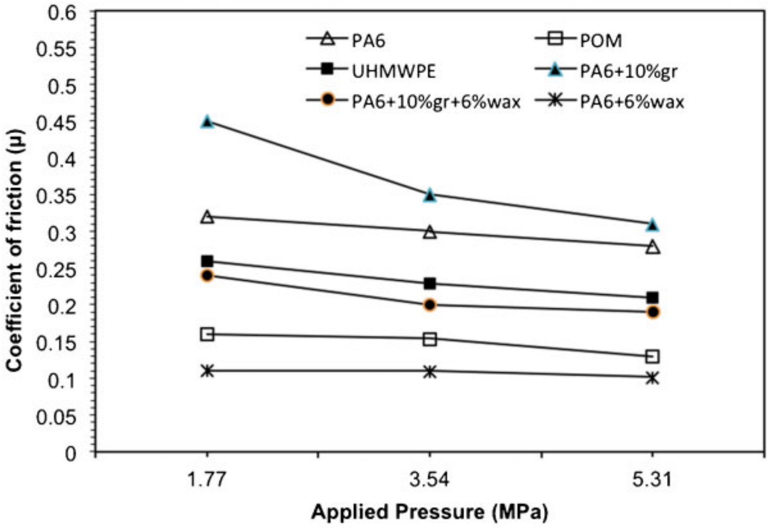
Figures 1 and 2 present the variation of coefficient of friction of PA6, UHMWPE, POM, PA6+10 wt-% graphite, PA6+10 wt-% graphite+6 wt-% wax and PA6+6 wt-% wax polymers and polymer composites with applied pressure and sliding speed respectively. It is clear from Fig. 1 that the friction coefficient values for PA6, POM, PA6+10 wt-% graphite, PA6+10 wt-% graphite+6 wt-% wax and PA6+6 wt-% wax polymers and polymer composites almost decrease with the increase in applied pressure values. This result is in agreement with Refs. 25 and 26 and in disagreement with Ref. 27. This is because the friction performance differs and is dependent on the applied load or applied pressure values range.[6,7,15] This figure also shows that the highest friction coefficient μ is 0·45 for PA6+10 wt-% graphite, and the lowest is μ=0·1 for PA6+6 wt-% wax composites. Figure 2 shows that the coefficient of friction of POM and PA6+6 wt-% wax composite is

Table 1 Polymer and polymer composite material's physical and mechanical properties

Materials	UHMWPE	POM	PA6	PA6+10% graphite	PA6+6% wax	PA6+10% graphite+ 6% wax
Properties						
Density/g cm^{-3}	0·93	1·41	1·13	1·19	1·11	1·16
Tensile strength at break/MPa	40	64	48	69	61	58
Tensile modulus/MPa	689	2700	2820	3400	2443	3428
Tensile strain at break/%	300	30	7·09	7·5	30	6·47
Izod impact strength/kJ m^{-2}	Non-break	7·0	7·8	7·5	8·6	7·5
Hardness/Shore D	66	80 HR$_M$	75±1	78±1	75±1	78±1
Suppliers	Tivar1000-Quadrant	Hostaform-Ticona	Domamid-Domopolymer			

Table 2 Specific test conditions of samples

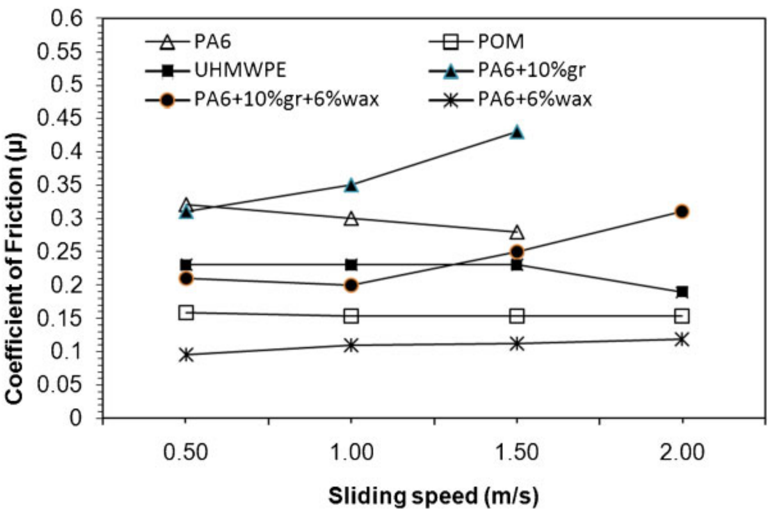
Materials	Temperature/°C	Humidity/%	Applied load/N	Sliding speed/m s^{-1}
UHMWPE				
POM	…	…	50	0·5
PA6	21±2	50±7	100	1·0
PA6+10% graphite	…	…	150	2·0
PA6+6% wax	…	…	…	…
PA6+10% graphite+ 6% wax	…	…	…	…



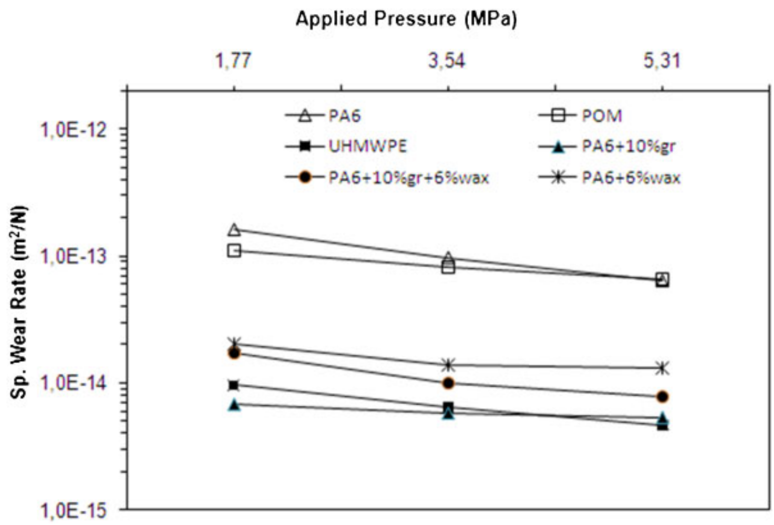
1 Friction coefficient–applied pressure curves for polymer and polymer composites against AISI 4140 steel at 1·0 m s^{-1} sliding speed

insensitive to the increase in sliding speed, while the rest of the tested polymers and polymer composite friction behaviour are following an increasing or decreasing trend with the increasing sliding speed values. The PA6+10 wt-% graphite and PA6+10 wt-% graphite+6 wt-% wax composites showed an increasing trend while PA6 and UHMWPE are following a decreasing trend with increasing sliding speed. The friction coefficient performances of the tested materials in Figs. 1 and 2 could be explained as follows. First, for polymer composites, the addition of graphite into PA6 could cause the graphite to locate and agglomerate between polyamide molecules, which decreases the bonding force of the polyamide molecules, hence to less surface bonding force of the material. Second, due to the temperature rises during the rubbing process, which cause a softening of the rubbed surfaces,[28] both of these reasons result in a soft and weak composite, which leads to the higher coefficient of friction of the composites than the neat polyamide.[29] Incorporating wax with graphite into PA6 plays the role of a sliding lubrication media, which compensates the negative role of the graphite and tends to lower friction coefficient values for PA6 composite than that of the neat polyamide.[30]

Figures 3 and 4 illustrate the variation of the specific wear rates of PA6, UHMWPE , POM, PA6+10 wt-% graphite, PA6+10 wt-% graphite+6 wt-% wax and PA6+6 wt-% wax polymers and polymer composites with applied pressure and sliding speed values respectively. Figure 3 shows that the specific wear rates decrease with increasing applied pressure values, while Fig. 4 shows that the specific wear rates increase with increasing sliding speed values. Among all the tested polymers and polymer composites, the lowest wear rates are for PA6+10 wt-% graphite composite and UHMWPE. This is followed by PA6+10 wt-% graphite+6 wt-% wax, PA6+6 wt-% wax, POM and PA6 materials respectively. Apart from UHMWPE, the wear rate of neat polymers is higher than that of polymer composites. This behaviour is due to the softening and melting of the worn surface of neat polymer, which pose a lower melting temperature than the composites.[29,31] Furthermore, the presence of protective film on the composite materials and UHMWPE worn surfaces plays a main role in lowering the wear rate of these materials.[23,31,32] The presence of the film on UHMWE is expected because of its very low glassy temperature



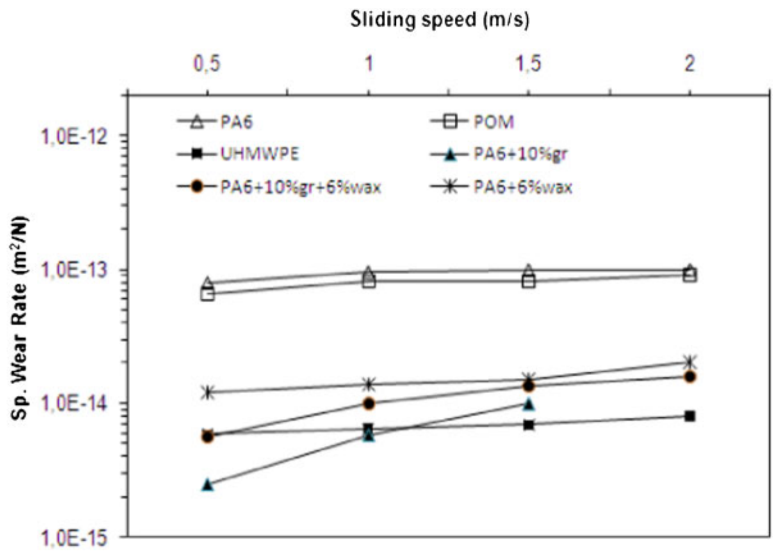
2 Friction coefficient–sliding speed curves for polymer and polymer composites against AISI 4140 steel at 3·54 MPa



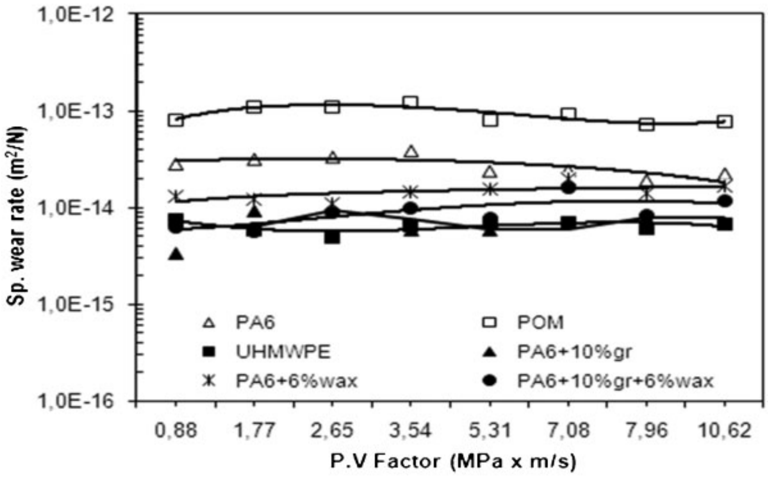
3 Specific wear rate–applied pressure curves for polymer and polymer composites against AISI 4140 steel at 1·0 m s^{-1} sliding speed

value (−120°C). Furthermore, the presence of wax filler in the composites places a lubricating media that protect the rubbed surfaces and result in the lower specific wear rate values of these composites.[28] Figure 5 presents the specific wear rate PV factor (combined pressure and sliding speed) curves for PA6, UHMWPE , POM, PA6+10 wt-% graphite, PA6+10 wt-% graphite+6 wt-% wax and PA6+6 wt-% wax polymers and polymer composites. It is clear from this figure that the wear rate is almost not influenced by the increasing PV factor values. This could be explained by the opposite influence of applied pressure and sliding speed on the specific wear rate (*see* Figs. 3 and 4). Hence, the applied pressure and sliding speed compensate each other influences in their combined factor PV. This is in agreement with Ref. 15. It is also clear from Fig. 5 that the specific wear rates of UHMPE and PA6+10 wt-% graphite composite are in the order of 10^{-15}. The specific wear rates of PA6+6 wt-% wax composites are in the order of 10^{-14}, while the neat POM and PA6 wear rate values are in the order of 10^{-13}. The wear rates of UHMWPE and PA6+10 wt-% graphite composite are 3400% lower than that of unfilled PA6 polymer. Although the wear rate of UWHMPE is close to PA6+10 wt-% graphite composite and presents a good tribological performance, the processing and manufacturing costs of PA6 composite are lower than that of UHMWPE polymer. Therefore, the PA6+10 wt-% graphite composite is preferred.

Figures 6 and 7 present the scanning and optical microscopy of the worn surfaces of PA6, UHMWPE, POM, PA6+10 wt-% graphite, PA6+10 wt-% graphite+6 wt-% wax and PA6+6 wt-% wax polymers and polymer composite pins and the steel disc worn surfaces respectively. The presence of transfer film patches on PA6+10 wt-% graphite composite and UHMWPE worn surfaces is clear (*see* Figs. 6*a* and *b* and 7*a* and *b*). Therefore, this transfer film has its major role in governing the wear resistance of the these polymer and polymer composites.[32] Figure 6*c* and *d* shows the presence of the deformation process, and Fig. 7*c* and *d* shows the presence of wax patches and transfer film on the worn surfaces. This shows the presence of deformation and transfer film mechanisms during the rubbing process. Furthermore, it is clear from Fig. 6*f* that the

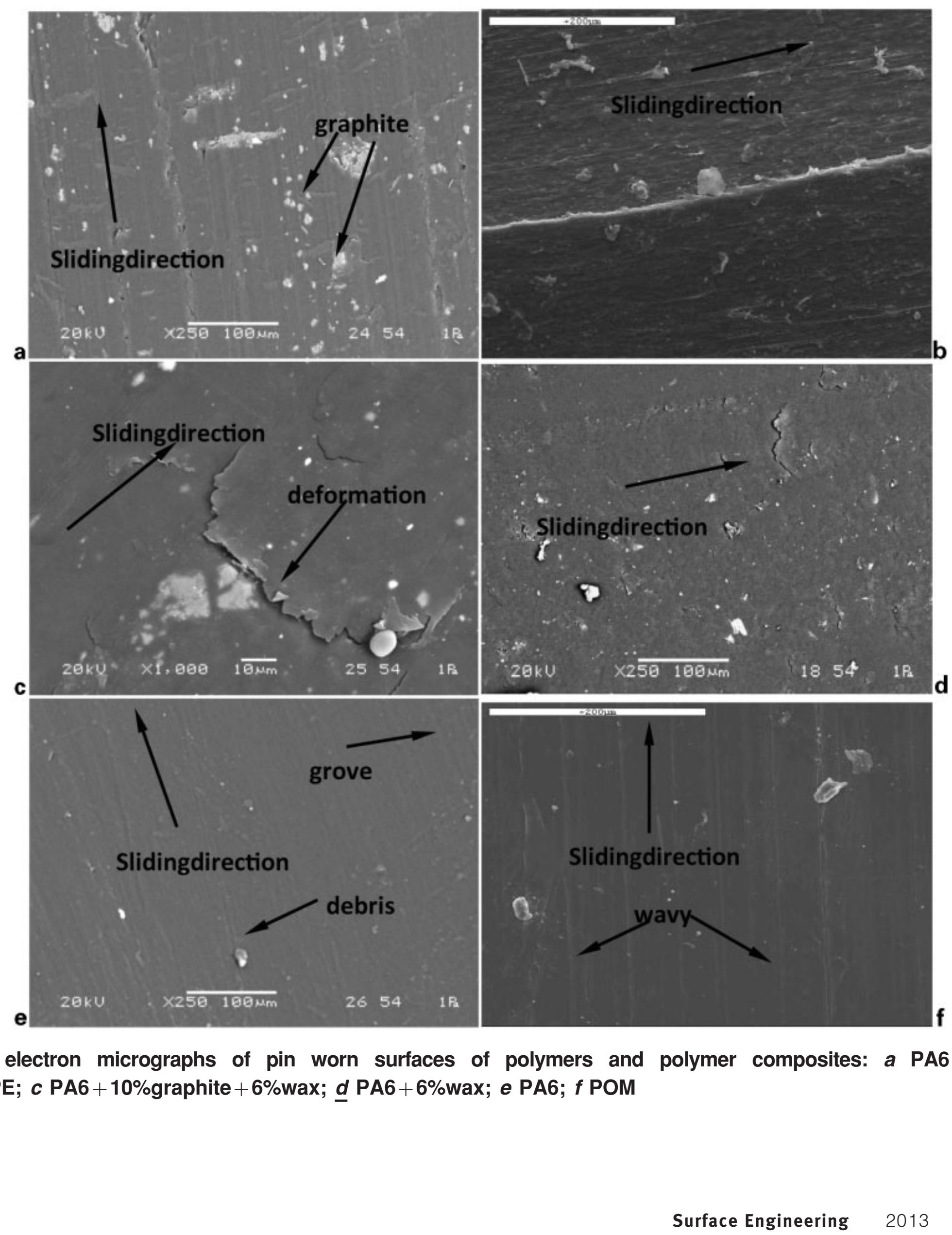


4 Specific wear rate–sliding speed curves for polymer and polymer composites against AISI 4140 steel at 3·54 MPa

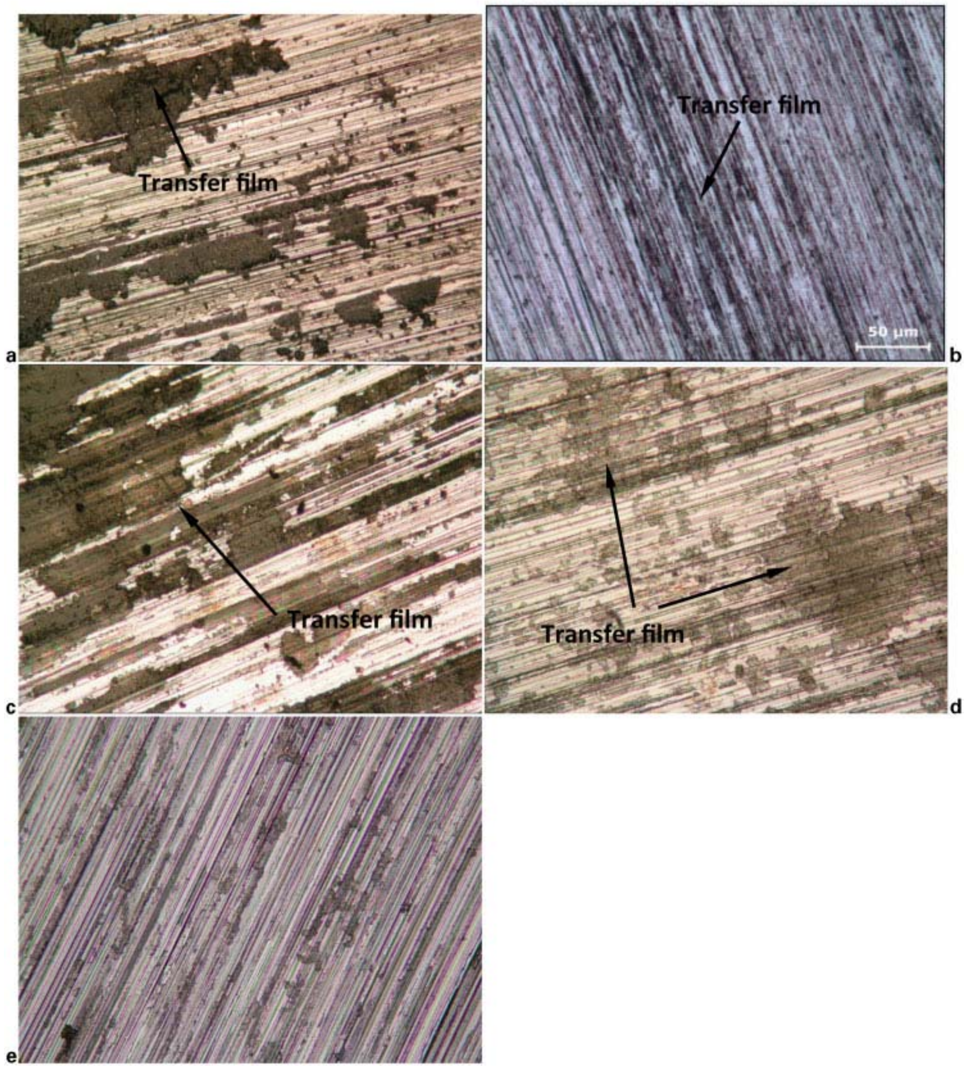


5 Specific wear rate–PV factor value curves for polymer and polymer composites under dry sliding condition

worn surface of POM is wavy, which is a result of the thermal effect due to the rise in temperature during the rubbing process. Hence, to a high level of specific wear rates. Figures 6*e* and 7*e* show the presence of grooves on the worn surface of neat PA6 polymer. These grooves have resulted from the presence of free PA6 debris between rubbed surfaces, which again lead to high level specific wear rates. Finally, the worn surface microscopy of the worn surfaces indicates the formation of transfer film, deformation and adhesive wear mechanisms.



6 Scanning electron micrographs of pin worn surfaces of polymers and polymer composites: *a* PA6+10%graphite; *b* UHMWPE; *c* PA6+10%graphite+6%wax; *d* PA6+6%wax; *e* PA6; *f* POM



7 Optical photographs of disc worn surfaces against polymers and polymer composites: *a* PA6+10%graphite; *b* UHMWPE; *c* PA6+10%graphite+6%wax; *d* PA6+6%wax; *e* PA6

Conclusion

1. For the tested materials and for the range of test condition of this investigation, the highest friction coefficient μ is 0·45 for PA6+10 wt-% graphite and the lowest is μ=0·1 for PA6+6 wt-% wax composites.

2. Within the range of test conditions of this investigation, the specific wear rates of PA6, UHMWPE , POM, PA6+10 wt-% graphite, PA6+10 wt-% graphite+6 wt-% wax and PA6+6 wt-% wax polymers and polymer composites are insignificantly influenced by the change in PV factor values (combined speed and load values).

3. For the tested materials and the range of test condition of this investigation, the specific wear rates of UHMPE and PA6+10 wt-% graphite composite are in the order of 10^{-15}. The specific wear rates of PA6+6 wt-% wax composites are in the order of 10^{-14}, while neat POM and PA6 wear rate values are in the order of 10^{-13}.

4. The addition of graphite and wax to PA6 polymer enhanced its tribological performance.

5. Among all the tested neat polymers and polymer composites of this investigation, PA6+10 wt-% graphite composite and UHMWPE neat polymer showed the highest tribological performance, but PA6+10 wt-% graphite composite has low processing; hence, it is preferred and suggested for application at the high speed environment and with low processing cost.

6. The wear mechanisms include transfer film, deformation and adhesive mechanisms.

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